Internship Report

Mandelstam-Tamm Quantum Speed Limit in a Two Level System

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Abstract

This internship focused on understanding Quantum Speed Limit (QSL), which define the fundamental limits on the time required for a quantum system to evolve from one state to another. The project explored formulation of the Mandelstam-Tamm bound, and their applications in two state quantum systems.

Contents

Chapter 1

Introduction

This chapter introduces the background of the internship, the motivation behind the project, and the main objectives that were set out to be achieved.

1.1 Background

Quantum mechanics places fundamental constraints on the dynamics of physical systems. Determining the minimum time required for a quantum system to evolve into a distinguishably different state is a fundamental question that has direct implications for the ultimate speed of quantum computation and the fundamental limits of quantum control.

The energy-time uncertainty relation $\Delta H \Delta t \sim \hbar$ has long been invoked as an answer, but its interpretation is subtle. A rigorous formulation was first provided by Mandelstam and Tamm in 1945, who derived a precise inequality providing a bound on how fast any dynamical quantity can change in terms of the system's energy uncertainty.

This internship, carried out at SRM University AP under Dr. Swetamber P. Das, focused on understanding the Quantum Speed Limit from first principles. The study began with a review of the quantum mechanical formalism before building up to the derivation of the Mandelstam-Tamm bound and its physical applications.

1.2 Objectives

The primary objectives of this internship were:

- To derive the Mandelstam-Tamm quantum speed limit.
- To generalize the MT bound to the case of non-orthogonal initial and final states.
- To apply the MT bound explicitly to a two-state quantum system governed by the Hamiltonian $H = \frac{\hbar\omega_0}{2}\sigma_z$, computing the energy uncertainty, the QSL time, and the time-dependent angle and visualize the resulting time evolution computationally

Chapter 2

Theory

This chapter establishes the theoretical foundation for the Quantum Speed Limit, beginning with the historical context of the energy-time uncertainty principle and culminating in the rigorous derivation of the Mandelstam-Tamm bound.

2.1 The Energy-Time Uncertainty Principle

In standard quantum mechanics, the Heisenberg uncertainty principle places fundamental limits on the precision with which certain pairs of conjugate variables can be simultaneously known (e.g., $\Delta x \Delta p \geq \hbar/2$) [3].

Historically, a similar relation for energy and time has been widely used:

$$\Delta E \Delta t \geq \frac{\hbar}{2} \tag{2.1}$$

While practically useful, its theoretical interpretation fundamentally differs from the position-momentum uncertainty relation. In standard quantum mechanics, time t is not an observable represented by a Hermitian operator; it is an external, continuous parameter tracking dynamical evolution [3]. Consequently, Δt does not represent the statistical dispersion of multiple measurements. Instead, Δt represents a characteristic time scale: it is the minimum time required for the expectation value of some dynamical observable to change by an amount equal to its standard deviation.

To formalize this interpretation and apply it to state evolution, Mandelstam and Tamm (1945) circumvented the lack of a time operator by anchoring the concept of "time uncertainty" directly to the survival probability of the system [4]. They derived a rigorous lower bound on the time required for an initial quantum state to evolve into a distinguishable, orthogonal state, relying purely on the well-established Robertson uncertainty relation [2].

2.2 Tools and Technologies

The computational aspects of this project were purely focused on graphing and visualizing the theoretical bound. For this purpose, the Python programming language was utilized, specifically relying on the `numpy` library for numerical calculations and arrays, and the `matplotlib` library for generating the plots of the time evolution and Bures angle.

2.3 Derivation of the Mandelstam-Tamm Bound

The core of this project centers on rigorously deriving the Mandelstam-Tamm (MT) Quantum Speed Limit from first principles [2].

2.3.1 The Robertson Uncertainty Relation and Density Matrices

We begin with the generalized Robertson uncertainty relation for two observables, H (the Hamiltonian) and A:

$$\Delta H \Delta A \geq \frac{1}{2}|\langle [H, A] \rangle| \tag{2.2}$$

To find a bound on the time evolution of the system, we need to relate the commutator to the time derivative of an observable. The time evolution of the expectation value of any operator A (that does not explicitly depend on time) is governed by the Heisenberg equation of motion, which is intimately connected to the von Neumann equation:

$$\frac{d\langle A \rangle}{dt} = \frac{i}{\hbar}\langle [H, A] \rangle \implies |\langle [H, A] \rangle| = \hbar \left| \frac{d\langle A \rangle}{dt} \right| \tag{2.3}$$

Substituting this into the uncertainty relation yields:

$$\Delta H \Delta A \geq \frac{\hbar}{2} \left| \frac{d\langle A \rangle}{dt} \right| \tag{2.4}$$

To measure how fast the system leaves its initial state $|\psi(0)\rangle$, we strategically choose the observable A to be the projection operator onto the initial state: $A = |\psi(0)\rangle\langle\psi(0)|$. This operator is mathematically equivalent to the density matrix of the initial pure state ($\rho(0)$).

Density matrices are fundamentally involved in this formulation because they naturally encapsulate the statistical state of a quantum system, and their time evolution is governed directly by the von Neumann equation (the quantum analogue of Liouville's theorem in classical mechanics) [1]. A crucial property of the density matrix for a pure state (or any projection operator) is idempotency, meaning $A^2 = A$.

Using this property, the variance of A simplifies significantly:

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2} = \sqrt{\langle A \rangle - \langle A \rangle^2} \tag{2.5}$$

The expectation value $\langle A \rangle_t = \langle \psi(t)|A|\psi(t)\rangle = |\langle \psi(0)|\psi(t)\rangle|^2$ represents the survival probability (or fidelity) of the initial state at time t.

2.3.2 Integrating the Bound for Orthogonal States

Substituting our simplified ΔA into the uncertainty relation gives:

$$\Delta H \sqrt{\langle A \rangle - \langle A \rangle^2} \geq \frac{\hbar}{2} \left| \frac{d\langle A \rangle}{dt} \right| \tag{2.6}$$

As the quantum system evolves away from its initial state, the survival probability $\langle A \rangle$ decreases from its initial value of 1, meaning $\frac{d\langle A \rangle}{dt} \leq 0$. Thus, $\left| \frac{d\langle A \rangle}{dt} \right| = -\frac{d\langle A \rangle}{dt}$. Separating variables and integrating from $t = 0$ (where $\langle A \rangle_0 = 1$) to time t:

$$\int_0^t \frac{2\Delta H}{\hbar} dt \geq \int_1^{\langle A \rangle_t} \frac{-d\langle A \rangle}{\sqrt{\langle A \rangle (1 - \langle A \rangle)}} \tag{2.7}$$

By substituting $\langle A \rangle = \sin^2 \theta$, the integral evaluates exactly to:

$$\frac{2\Delta H t}{\hbar} \geq \pi - 2 \arcsin(\sqrt{\langle A \rangle_t}) \implies \frac{\Delta H t}{\hbar} \geq \frac{\pi}{2} - \arcsin(\sqrt{\langle A \rangle_t}) \tag{2.8}$$

To find the minimum time required for the system to reach a completely *orthogonal* state, we set $\langle \psi(0)|\psi(\tau)\rangle = 0$, which means $\langle A \rangle_\tau = 0$.

$$\frac{\Delta H \tau_{QSL}}{\hbar} \geq \frac{\pi}{2} - \arcsin(0) = \frac{\pi}{2} \tag{2.9}$$

Rearranging for τ_{QSL}, we recover the famous Mandelstam-Tamm Quantum Speed Limit for orthogonal states:

$$\tau_{QSL} \geq \frac{\pi \hbar}{2\Delta H} \tag{2.10}$$

2.4 Generalized Mandelstam-Tamm Bound

In many practical scenarios, a quantum system might not evolve to a completely orthogonal state, but rather to a target state separated by some Bures angle Θ. We can define the inner product overlap in terms of this angle as $\langle A \rangle_t = \cos^2 \Theta$.

Plugging this non-orthogonal overlap into our integrated bound gives:

$$\begin{aligned} \frac{\Delta H t}{\hbar} &\geq \frac{\pi}{2} - \arcsin\left(\sqrt{\cos^2\Theta}\right) \\ &= \frac{\pi}{2} - \arcsin(\cos\Theta) \\ &= \frac{\pi}{2} - \left(\frac{\pi}{2} - \Theta\right) = \Theta \end{aligned} \tag{2.11}$$

This provides the generalized Mandelstam-Tamm bound for any arbitrary angle Θ:

$$t \geq \frac{\hbar\Theta}{\Delta H} \tag{2.12}$$

This generalized formulation elegantly ensures that the time required to traverse an angle Θ in the Hilbert space is fundamentally restricted by the energy uncertainty of the system, bridging the gap between orthogonal and non-orthogonal state evolution.

Chapter 3

Application to a Two-State System

This chapter applies the theoretical framework of Mandelstam and Tamm's Quantum Speed Limit discussed in the methodology to a specific problem: a two-state quantum system.

3.1 System Hamiltonian and Time Evolution

Consider a two-state quantum system governed by the Hamiltonian:

$$H = \frac{\hbar\omega_0}{2}\sigma_z = \frac{\hbar\omega_0}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.1}$$

where σ_z is the Pauli Z matrix. The unitary time evolution operator is given by:

$$U(t) = e^{-iHt/\hbar} = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix} \tag{3.2}$$

3.2 Energy Variance and the MT Bound

To apply the Mandelstam-Tamm bound, we first compute the energy variance ΔH of the initial state. The expectation value of the Hamiltonian is:

$$\langle H \rangle = \langle \psi(0)|H|\psi(0)\rangle = \frac{\hbar\omega_0}{2}\left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right) = \frac{\hbar\omega_0}{2}\cos\theta \tag{3.3}$$

Since $H^2 = \left(\frac{\hbar\omega_0}{2}\right)^2 I$, the expectation value of H^2 is simply $\langle H^2\rangle = \left(\frac{\hbar\omega_0}{2}\right)^2$. The energy fluctuation (standard deviation) is therefore:

$$\Delta H = \sqrt{\langle H^2\rangle - \langle H\rangle^2} = \sqrt{\left(\frac{\hbar\omega_0}{2}\right)^2 - \left(\frac{\hbar\omega_0}{2}\cos\theta\right)^2} = \frac{\hbar\omega_0}{2}\sin\theta \tag{3.4}$$

According to the Mandelstam-Tamm bound, the time t required for the system to evolve to a state separated by a Bures angle Θ from the initial state is bounded by:

$$t \geq \frac{\hbar\Theta}{\Delta H} = \frac{\hbar\Theta}{\frac{\hbar\omega_0}{2}\sin\theta} = \frac{2\Theta}{\omega_0 \sin\theta} \tag{3.5}$$

For the system to reach an orthogonal state, the distance is $\Theta = \pi/2$. The minimum time required to reach this orthogonal state is thus:

$$t \geq \frac{2\cdot(\pi/2)}{\omega_0\sin\theta} \implies \tau_{QSL} = \frac{\pi}{\omega_0\sin\theta} \tag{3.6}$$

This indicates that the speed of evolution depends on the initial state parameter θ. The fastest evolution to an orthogonal state occurs when $\theta = \pi/2$ (states on the equator of the Bloch sphere), yielding a minimum time of $\tau_{QSL} = \pi/\omega_0$.

3.3 Generalized Mandelstam-Tamm Bound

Let the initial state of the system be parameterized by an angle θ on the Bloch sphere:

$$|\psi(0)\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle \tag{3.7}$$

The state at a later time t is:

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle = \cos\frac{\theta}{2}e^{-i\omega_0 t/2}|0\rangle + \sin\frac{\theta}{2}e^{i\omega_0 t/2}|1\rangle \tag{3.8}$$

The inner product between the initial and evolved state is:

$$\langle\psi(0)|\psi(t)\rangle = \cos^2\frac{\theta}{2}e^{-i\omega_0 t/2} + \sin^2\frac{\theta}{2}e^{i\omega_0 t/2} = \cos\left(\frac{\omega_0 t}{2}\right) - i\cos\theta\sin\left(\frac{\omega_0 t}{2}\right) \tag{3.9}$$

The squared magnitude of this overlap gives the fidelity:

$$|\langle\psi(0)|\psi(t)\rangle|^2 = \cos^2\left(\frac{\omega_0 t}{2}\right) + \cos^2\theta\sin^2\left(\frac{\omega_0 t}{2}\right) \tag{3.10}$$

The Bures angle $\alpha(t)$ between the initial and evolved state is defined as:

$$\alpha(t,\theta) = \arccos\left(|\langle\psi(0)|\psi(t)\rangle|\right) = \arccos\sqrt{\cos^2\left(\frac{\omega_0 t}{2}\right) + \cos^2\theta\sin^2\left(\frac{\omega_0 t}{2}\right)} \tag{3.11}$$

3.4 Computational Analysis

Using Python, we computationally plotted the Bures angle $\alpha(t)$ as a function of the dimensionless time $\omega_0 t$ for various initial states (different values of θ) to analyze the generalized Mandelstam-Tamm (MT) bound.

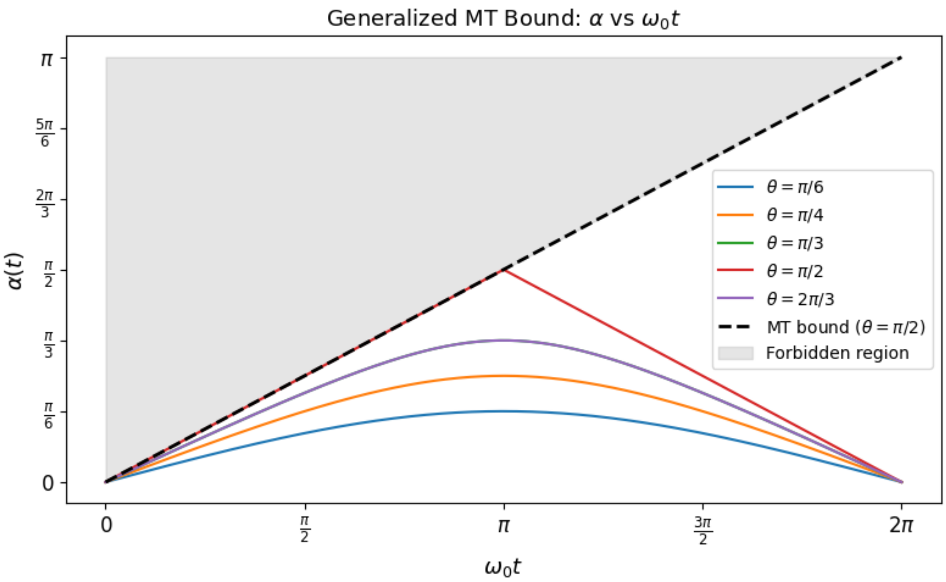

Figure 3.1: Generalized Mandelstam-Tamm Bound: $\alpha(t)$ vs $\omega_0 t$ for different initial states θ. The black dashed line represents the MT bound for orthogonal states ($\theta = \pi/2$), and the shaded gray region is forbidden by the bound.

As observed in Figure 3.1, for all initial states, the time evolution strictly respects the generalized MT bound. The evolution is fastest for $\theta = \pi/2$ (corresponding to states on the equator of the Bloch sphere), which reaches an orthogonal state ($\alpha = \pi/2$) at $\omega_0 t = \pi$. For other values of θ, the system never reaches an orthogonal state, and the curve stays below the theoretical maximum bound.

Chapter 4

Results and Conclusion

This chapter summarizes the findings of the internship project, discusses the physical implications of the derived bound, and outlines potential directions for future study.

4.1 Summary

This project successfully explored the fundamental limit of quantum dynamics. We began by reviewing the underlying quantum mechanical formalism, establishing a solid theoretical foundation in state vectors, operators, and density matrices. With this framework, we rigorously derived the Mandelstam-Tamm (MT) Quantum Speed Limit from the Robertson uncertainty relation, bypassing the historical ambiguities of the energy-time uncertainty principle.

The bound was then generalized to account for the evolution between non-orthogonal initial and final states. Finally, we explicitly applied this generalized bound to a two-level quantum system and verified it computationally. The numerical simulations successfully confirmed that the speed of the system's state evolution is strictly bounded by its energy variance, with the fastest possible evolution occurring for initial states with maximum energy uncertainty (states on the equator of the Bloch sphere).

4.2 Discussion and Physical Meaning

The Mandelstam-Tamm bound holds profound implications for the physical reality of dynamic systems. It dictates that quantum states possess a fundamental, inescapable "speed limit" that is entirely governed by their energy variance (ΔH).

In practical terms, this places a hard physical limit on how fast quantum processes can occur. For instance, in the realm of quantum computing, the clock speed of a quantum processor—the minimum time required to reliably flip a qubit from $|0\rangle$ to $|1\rangle$ (an orthogonal state)—cannot be made arbitrarily short. It is strictly bounded by the energy resources (specifically, the energy spread) available to the system. A faster operation inherently requires a larger energy variance. This means that designing ultra-fast quantum computers or highly efficient quantum thermodynamic engines will eventually hit a fundamental physical bottleneck dictated by the MT bound, representing the absolute theoretical maximum of quantum control.

4.3 Future Work

While this internship focused deeply on the Mandelstam-Tamm formulation, the study of the Quantum Speed Limit is a vast and active field of research. Future work could build upon this foundation in several exciting directions:

- **The Margolus-Levitin Bound:** Exploring the other primary quantum speed limit, the Margolus-Levitin (ML) bound. While the MT bound restricts evolution based on energy variance (ΔH), the ML bound restricts it based on the mean energy above the ground state ($E - E_0$). Investigating the crossover regime where one bound becomes tighter than the other would provide a complete picture of pure state dynamics.

- **Open Quantum Systems:** The current study assumed a closed quantum system undergoing isolated, unitary time evolution. A natural next step would be to apply the QSL to open quantum systems, where the system interacts with an external environment, leading to decoherence and non-unitary dynamics.

- **Many-Body Systems:** Extending the application of this bound from a simple two-state system to complex many-body quantum systems, to study how entanglement and quantum correlations affect the overall speed of evolution.

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